A Reduced Model for Bragg-Interferometer Phase Noise Induced by Raman Velocity-Selection Noise

Project note for the local Python analysis tool

Abstract

We document a reduced calculation model for estimating how shot-to-shot fluctuations of the Raman-selected velocity class propagate into the phase of a first-order Bragg Mach–Zehnder atom interferometer. The implementation targets the current ^{87}Rb configuration with a $\pi/2$–π–$\pi/2$ pulse sequence, Gaussian Bragg pulses, and a Gaussian velocity-selected ensemble. Two observables are emphasized: the total interferometer fringe phase and an operational pulse-only Bragg diffraction phase obtained by suppressing dark-time phase accumulation while keeping the same finite-duration pulses. The document also defines the exact quantities shown in the four panels of the desktop interface: selected velocity distributions, phase-versus-center curves, phase-noise-versus-Raman-noise curves, and contrast-versus-center curves.

1 Introduction and Scope

The physical question is the following: after Raman velocity selection, each experimental shot prepares a narrow but fluctuating velocity class. Even if the microscopic origin of the fluctuation is not modeled, the resulting shot-to-shot variation of the selected mean velocity changes the effective Doppler detuning seen by the Bragg pulses. Because finite-duration Bragg pulses have detuning-dependent transfer amplitudes and phases, the mean velocity fluctuation is converted into a fluctuation of the interferometer output phase.

The present implementation is intentionally restricted to a controlled first model:

- atomic species: ^{87}Rb,
- interferometer sequence: $\pi/2$–π–$\pi/2$,
- Bragg order: first order,
- Bragg pulses: Gaussian envelopes in an effective two-level model,
- selected atoms: one-dimensional Gaussian detuning distribution,
- shot-to-shot Raman noise: Gaussian noise on the center detuning of that selected distribution,
- cloud size, spatial inhomogeneity, and position–velocity correlations: neglected.

This scope is sufficient to quantify, within a simplified but internally consistent model, how a Raman center-noise RMS maps to a Bragg-induced phase-noise RMS.

2 Default Experimental Configuration

The default user-interface values correspond to the current experimental discussion:

Quantity	Default value
Atomic species	^{87}Rb
Bragg order	$n = 1$
Pulse sequence	$\pi/2$–π–$\pi/2$
Beam-splitter duration	25 µs
Mirror duration	50 µs
Bragg pulse shape	Gaussian
Dark time	8 ms
Selected-distribution FWHM	12 kHz
Reference Raman center-noise RMS	4 kHz
Nominal center detuning	0 kHz relative to Bragg resonance

The statement that the Raman-selected $+\hbar k$ class corresponds to approximately 15 kHz in the Raman configuration is treated here as calibration context. In the current Bragg solver, the dynamical variable is the detuning relative to the nominal Bragg resonance of the selected class.

3 Effective Bragg Pulse Model

3.1 Two-level Hamiltonian

For a given Doppler-equivalent two-photon detuning δ and pulse phase φ, the first-order Bragg pulse is modeled in the basis

$$\{\, |0\rangle,\ |2\hbar k\rangle \,\}$$

by the effective Hamiltonian

$$\frac{H(t;\delta,\varphi)}{\hbar} = \frac{1}{2}\begin{pmatrix} -\delta & \Omega(t)e^{-i\varphi} \\ \Omega(t)e^{+i\varphi} & \delta \end{pmatrix}. \tag{1}$$

For the first two pulses the code uses $\varphi = 0$, while the final pulse phase is the scanned interferometer phase.

3.2 Gaussian pulse envelope and area calibration

The Bragg envelope is written as

$$\Omega(t) = \Omega_0 \exp\left(-\frac{t^2}{2\tau^2}\right), \tag{2}$$

where τ is the RMS temporal width used internally by the solver. Since the UI may specify the pulse duration under different conventions, the code first converts the user entry to τ.

The pulse is truncated numerically at $|t| \le N_\sigma \tau$. The peak Rabi frequency Ω_0 is set by the target pulse area,

$$\theta = \int_{-N_\sigma\tau}^{+N_\sigma\tau} \Omega(t)\,dt = \Omega_0 \tau \sqrt{2\pi}\,\mathrm{erf}\left(\frac{N_\sigma}{\sqrt{2}}\right), \tag{3}$$

with $\theta = \pi/2$ for the beam splitter and $\theta = \pi$ for the mirror.

3.3 Time-discretized exact propagator

For each short time step Δt the Hamiltonian is approximated as constant, and the corresponding propagator is evaluated analytically. Writing

$$M(\delta,\Omega) = \begin{pmatrix} -\delta & \Omega \\ \Omega & \delta \end{pmatrix}, \tag{4}$$

with phase absorbed into the chosen basis for the relevant step, the step propagator is

$$U_{\text{step}}(\delta, \Omega, \Delta t) = \cos\left(\frac{\Omega_{\text{eff}} \Delta t}{2}\right) I - i \frac{\sin\left(\frac{\Omega_{\text{eff}} \Delta t}{2}\right)}{\Omega_{\text{eff}}} M(\delta, \Omega), \tag{5}$$

where

$$\Omega_{\text{eff}} = \sqrt{\delta^2 + \Omega^2}. \tag{6}$$

The unitary of a finite Gaussian pulse is obtained by ordered multiplication of Eq. (5) along the pulse.

4 Free Evolution and Pulse Sequence

In the same rotating-frame description, dark-time evolution is represented by

$$U_{\text{free}}(T) = \begin{pmatrix} e^{+i\delta T/2} & 0 \\ 0 & e^{-i\delta T/2} \end{pmatrix}. \tag{7}$$

For a fixed detuning class Δ, the full three-pulse sequence is therefore

$$U_{\text{MZ}}(\Delta, \phi) = U_{\text{BS}}(\Delta, \phi) U_{\text{free}}(T) U_{\text{M}}(\Delta) U_{\text{free}}(T) U_{\text{BS}}(\Delta), \tag{8}$$

where ϕ is the phase of the final beam splitter.

5 Raman-Selected Ensemble and Shot-to-Shot Noise

The Raman-selected atoms are represented by a Gaussian detuning distribution

$$g(\Delta; \mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{(\Delta - \mu)^2}{2\sigma^2}\right], \tag{9}$$

where μ is the shot-dependent center detuning. The selected-distribution width is parameterized by the experimental FWHM,

$$\sigma = \frac{\text{FWHM}}{2\sqrt{2 \ln 2}}. \tag{10}$$

The Raman center noise is then modeled as

$$\mu_{\text{shot}} \sim \mathcal{N}\left(\mu_0, \sigma_{\text{noise}}^2\right), \tag{11}$$

with nominal center μ_0 and shot-to-shot RMS σ_{noise}.

For ensemble observables, averages over Eq. (9) are evaluated numerically by Gauss–Hermite quadrature.

6 Fringe Coefficient, Phase, and Contrast

6.1 Single-detuning fringe coefficient

Let the input state be $|\psi_{\text{in}}\rangle = |0\rangle$. For a fixed detuning class Δ, the population in the $|0\rangle$ output port after the full sequence can always be written as

$$P_0(\phi \mid \Delta) = A(\Delta) + 2\,\text{Re}\left[c(\Delta) e^{i\phi}\right], \tag{12}$$

where $A(\Delta)$ is the phase-independent offset and $c(\Delta)$ is the complex fringe coefficient. The total single-class phase is therefore

$$\phi_{\text{tot}}(\Delta) = -\arg[c(\Delta)]. \tag{13}$$

6.2 Ensemble-averaged total phase

The ensemble-averaged fringe coefficient is

$$\bar{c}_{\mathrm{tot}}(\mu) = \int g(\Delta;\mu,\sigma)\, c_{\mathrm{tot}}(\Delta)\, d\Delta, \tag{14}$$

and the corresponding total interferometer phase is

$$\Phi_{\mathrm{tot}}(\mu) = -\arg[\bar{c}_{\mathrm{tot}}(\mu)]\,. \tag{15}$$

The associated ensemble contrast is defined by

$$C_{\mathrm{tot}}(\mu) = \frac{2\,|\bar{c}_{\mathrm{tot}}(\mu)|}{\bar{A}_{\mathrm{tot}}(\mu)}, \qquad \bar{A}_{\mathrm{tot}}(\mu) = \int g(\Delta;\mu,\sigma) A_{\mathrm{tot}}(\Delta)\, d\Delta. \tag{16}$$

6.3 Operational pulse-only diffraction phase

To isolate the phase generated by the finite-duration Bragg pulses themselves, the tool also defines a pulse-only sequence obtained by setting the free-evolution operator to the identity while keeping the same pulse unitaries. The pulse-only pre-final state is

$$|\psi_{\mathrm{pre}}^{(\mathrm{diff})}\rangle = U_{\mathrm{M}}(\Delta) U_{\mathrm{BS}}(\Delta)|0\rangle. \tag{17}$$

Repeating the same fringe-coefficient extraction gives

$$\bar{c}_{\mathrm{diff}}(\mu) = \int g(\Delta;\mu,\sigma)\, c_{\mathrm{diff}}(\Delta)\, d\Delta, \tag{18}$$

$$\Phi_{\mathrm{diff}}(\mu) = -\arg[\bar{c}_{\mathrm{diff}}(\mu)]\,, \tag{19}$$

$$C_{\mathrm{diff}}(\mu) = \frac{2\,|\bar{c}_{\mathrm{diff}}(\mu)|}{\bar{A}_{\mathrm{diff}}(\mu)}. \tag{20}$$

This quantity is model dependent, but it is well-defined within the present solver and directly measures the detuning-dependent phase shift produced by the Bragg pulses in the absence of dark-time phase accumulation.

7 Propagation of Raman Center Noise into Phase Noise

The tool reports two mappings from Raman center-noise RMS to phase-noise RMS.

7.1 Linearized propagation

At the nominal center μ_0, the small-noise estimate is

$$\sigma_{\phi,\mathrm{tot}}^{(\mathrm{lin})} \approx \left|\frac{d\Phi_{\mathrm{tot}}}{d\mu}\right|_{\mu=\mu_0} \sigma_{\mathrm{noise}}, \tag{21}$$

$$\sigma_{\phi,\mathrm{diff}}^{(\mathrm{lin})} \approx \left|\frac{d\Phi_{\mathrm{diff}}}{d\mu}\right|_{\mu=\mu_0} \sigma_{\mathrm{noise}}. \tag{22}$$

In the code, these slopes are computed numerically by a centered finite difference in μ.

7.2 Monte Carlo propagation

The non-linear estimate is obtained by sampling

$$\mu_j \sim \mathcal{N}(\mu_0, \sigma^2_{\text{noise}}), \qquad j = 1, \ldots, N_{\text{MC}}, \tag{23}$$

and evaluating the wrapped phase excursions relative to the nominal shot,

$$\delta\Phi_{\text{tot},j} = \arg\{\exp[i\,(\Phi_{\text{tot}}(\mu_j) - \Phi_{\text{tot}}(\mu_0))]\}\,, \tag{24}$$

$$\delta\Phi_{\text{diff},j} = \arg\{\exp[i\,(\Phi_{\text{diff}}(\mu_j) - \Phi_{\text{diff}}(\mu_0))]\}\,. \tag{25}$$

The corresponding RMS values are

$$\sigma^{(\text{MC})}_{\phi,\text{tot}} = \text{Std}[\delta\Phi_{\text{tot},j}]\,, \qquad \sigma^{(\text{MC})}_{\phi,\text{diff}} = \text{Std}[\delta\Phi_{\text{diff},j}]\,. \tag{26}$$

8 Interpretation of the Four Plots Produced by the UI

The graphical interface displays four panels. Their mathematical definitions are summarized here so that the figures can be discussed in a paper-like way.

8.1 Panel 1: selected velocity distribution

The upper-left panel shows the nominal and shifted selected distributions. The nominal profile is

$$I_{\text{nom}}(\Delta) \propto \exp\left[-\frac{(\Delta - \mu_0)^2}{2\sigma^2}\right], \tag{27}$$

while the reference shifted profile shown by the UI corresponds to a +1 RMS center shift,

$$I_{+1\sigma}(\Delta) \propto \exp\left[-\frac{(\Delta - \mu_0 - \sigma_{\text{noise}})^2}{2\sigma^2}\right]. \tag{28}$$

These curves are not themselves interferometer observables; they visualize the Raman-selected input ensemble and the meaning of the shot-to-shot center fluctuation.

8.2 Panel 2: phase change versus shot center detuning

The upper-right panel plots the phase change relative to the nominal center. For the total interferometer phase, the plotted quantity is

$$\Delta\Phi_{\text{tot}}(\mu) = \text{unwrap}[\Phi_{\text{tot}}(\mu)] - \text{unwrap}[\Phi_{\text{tot}}(\mu_0)]\,. \tag{29}$$

Likewise, for the pulse-only diffraction phase,

$$\Delta\Phi_{\text{diff}}(\mu) = \text{unwrap}[\Phi_{\text{diff}}(\mu)] - \text{unwrap}[\Phi_{\text{diff}}(\mu_0)]\,. \tag{30}$$

This panel directly shows how sensitive the interferometer phase is to shot-to-shot shifts of the selected mean velocity.

8.3 Panel 3: phase noise versus Raman center-noise amplitude

The lower-left panel maps the Raman center-noise RMS σ_{noise} to the resulting phase-noise RMS. Four curves are shown:

$$\sigma_{\phi,\text{tot}}^{(\text{lin})}(\sigma_{\text{noise}}) = \left|\frac{d\Phi_{\text{tot}}}{d\mu}\right|_{\mu_0} \sigma_{\text{noise}}, \tag{31}$$

$$\sigma_{\phi,\text{tot}}^{(\text{MC})}(\sigma_{\text{noise}}) = \text{Std}[\delta\Phi_{\text{tot},j}]\,, \tag{32}$$

$$\sigma_{\phi,\text{diff}}^{(\text{lin})}(\sigma_{\text{noise}}) = \left|\frac{d\Phi_{\text{diff}}}{d\mu}\right|_{\mu_0} \sigma_{\text{noise}}, \tag{33}$$

$$\sigma_{\phi,\text{diff}}^{(\text{MC})}(\sigma_{\text{noise}}) = \text{Std}[\delta\Phi_{\text{diff},j}]\,. \tag{34}$$

Agreement between linearized and Monte Carlo curves indicates a locally linear response around μ_0; deviation indicates a non-linear phase response over the noise scale being probed.

8.4 Panel 4: contrast versus shot center detuning

The lower-right panel plots the ensemble-averaged contrast versus the shot center detuning,

$$C_{\text{tot}}(\mu) = \frac{2|\bar{c}_{\text{tot}}(\mu)|}{\bar{A}_{\text{tot}}(\mu)}, \qquad C_{\text{diff}}(\mu) = \frac{2|\bar{c}_{\text{diff}}(\mu)|}{\bar{A}_{\text{diff}}(\mu)}. \tag{35}$$

This panel is important because phase sensitivity and contrast are coupled: a steep phase response accompanied by strong contrast loss may indicate that the selected distribution is moving into a spectrally unfavorable region of the Bragg response.

9 Discussion of What the Four Plots Mean Physically

Taken together, the four panels provide a compact diagnostic chain:

1. Panel 1 defines the input ensemble and the magnitude of the imposed shot-to-shot Raman fluctuation.
2. Panel 2 converts that input fluctuation into a phase-response curve in detuning space.
3. Panel 3 turns the local or non-linear phase response into a directly useful noise metric, namely σ_ϕ.
4. Panel 4 tracks whether the same detuning excursion also degrades contrast, which is often a signature of operating away from optimal Bragg resonance.

In a paper-style discussion, Panel 2 is the response function, Panel 3 is the propagated noise metric, and Panel 4 is the associated visibility penalty.

10 Assumptions and Limitations

The model is rigorous only within the reduced description above. It does *not* yet include:

- coupling to higher Bragg momentum orders,
- explicit multi-state diffraction phases,
- spatial intensity inhomogeneity and wavefront effects,
- initial cloud size or position distribution,

- Raman pulse dynamics or a microscopic model for the origin of the center noise,
- correlations between center fluctuations and width fluctuations.

Accordingly, the current results should be interpreted as a controlled first estimate of how a noisy Raman-selected mean velocity maps into total fringe phase noise and pulse-only diffraction phase noise.

11 Code Organization

The implementation used by the desktop application is organized as follows:

- `main.py`: launches the desktop UI,
- `bragg_phase_noise/config.py`: model configuration,
- `bragg_phase_noise/physics.py`: Gaussian pulses and finite-step propagators,
- `bragg_phase_noise/analysis.py`: ensemble averages, phases, contrasts, and noise propagation,
- `bragg_phase_noise/ui.py`: Tkinter + Matplotlib interface.